AAD - Mod4

Dynamic Programming

Principle of Optimality: Optimal solution to a dynamic optimisation problem can be found by combining optimal solutions to its subproblems.

Matrix chain multiplication

$A_1 \quad A_2 \quad A_3 \quad A_4$

$m \times n \quad n \times o \quad o \times p \quad p \times q$

$d_0 \quad d_1 \quad d_2 \quad d_3 \quad d_4$

$$C[i,j] = \min_{i \leq k < j} \{ C[i,k] + C[k+1,j] + d_{i-1} \times d_k \times d_j \}$$

m table consists of min. cost
S table consists of k values that give the min. cost.

m	1	2	3	4
1				← Answer lies here
2				
3				
4				

Put) after Ax

S	1	2	3	4
1				x
2				
3				
4				

$A_1 (\underbrace{A_2 A_3 A_4})$

↳ For putting paranthesis, find the value (2,4) in S table, x, & put it after Ax.

Floyd - Warshall

```
Algorithm Floyd Warshall (cost[][],n)
{
  for i=1 to n do
     for j=1 to n do
        D[i,j] = cost[i,j]
  for k=1 to n do
     for i=1 to n do
        for j=1 to n do
           D[i,j] = min{ D[i,j],
                         D[i,k] + D[k,j] }
  return D
}
```

Time complexity = $O(n^3)$

N-Queens: Placing N no. of queens on an n x n chess board such that neither of them can attack one another.

Travelling Salesman

1) Create adjacency matrix. Diagonals = ∞
2) Perform row reduction & column reduction, i.e., find min value in a row & subtract each row with respective min values. Then do the same on columns as well.
3) Add up total row & column reduction values ⇒ cost.
4) Create state space tree with node name & cost.
5) From next node, set row a & column b as ∞.
6) Cost = c(a,b) + cost ~~from~~ of immediate parent node + cost of current reduction.